

Ram K

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SUMMARY

SENIOR AI ENGINEER WITH 18+ YEARS OF ENGINEERING EXPERIENCE, SPECIALIZING IN GENERATIVE AI, RAG, AGENTIC AI, AND ML. SKILLED IN BUILDING ENTERPRISE AI PLATFORMS USING PYTHON, LANGCHAIN, LANGGRAPH, LLMs, VECTOR DATABASES, FASTAPI, AND CLOUD TECHNOLOGIES.

EDUCATION

MASTER OF COMPUTER APPLICATIONS (MCA)
MK University
2001

TECHNICAL SKILLS

PROGRAMMING LANGUAGES

Python • Java • SQL • JavaScript • Bash

AI / GENERATIVE AI

Agentic AI • LLMs • LangGraph • LangChain • RAG • Prompt Engineering • OpenAI • Azure OpenAI • Ollama

MACHINE LEARNING

Scikit-learn • Logistic Regression • Classification • Feature Engineering • Model Evaluation • ROC-AUC • Pandas • NumPy

VECTOR DATABASES

FAISS • Pinecone • ChromaDB • Embeddings • Vector Search

AI FRAMEWORKS

FastAPI • Flask • REST APIs • JSON • Pydantic

CLOUD & MLOPS

AWS (EC2, S3, Lambda, ECS, IAM, CloudWatch) • Azure OpenAI • Docker • Kubernetes • GitHub Actions • Jenkins • MLflow

DATA ENGINEERING

PostgreSQL • MongoDB • Redis • Kafka • ETL Pipelines

ARCHITECTURE

Agent Workflows • Multi-Agent Systems • LLM Orchestration • Inference Pipelines • Event-Driven Architecture • Microservices

DEVELOPER TOOLS

Git • GitHub • VS Code • Jupyter Notebook • Linux

EXPERIENCE

ERCOT | SENIOR AI ENGINEER

Apr 2025 – Present | Taylor, TX

- Designed and developed enterprise AI solutions using Python, FastAPI, LangGraph, LangChain, and RAG to automate payment operations and business workflows.
- Built multi-agent AI systems for intent classification, root cause analysis, semantic search, trend analysis, recommendations, and executive reporting.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using ChromaDB, Ollama, and local LLMs for secure enterprise knowledge retrieval.
- Built payment monitoring and reporting solutions using Python, APScheduler, and Pandas to improve operational visibility and reduce manual effort.

AMAZON ROBOTICS | SENIOR AI ENGINEER

Jun 2023 – Mar 2025 | North Reading, MA

- Designed and developed AI-powered robotics solutions using Python, FastAPI, LangGraph, and RAG to automate warehouse operations, robot diagnostics, and predictive maintenance.
- Built a multi-agent AI architecture for intent classification, fleet diagnostics, semantic search, predictive maintenance, task optimization, and operational reporting using LangGraph and local LLMs.
- Developed Retrieval-Augmented Generation (RAG) pipelines using ChromaDB, Ollama, and vector embeddings to enable intelligent search and conversational AI over robotics operational data.

BIOGEN | AI ENGINEER

Jun 2022 – Jun 2023 | Durham, NC

- Integrated Llama-based Large Language Models (LLMs) with enterprise applications using FastAPI REST APIs to enable document intelligence and conversational AI.
- Designed and implemented Retrieval-Augmented Generation (RAG) pipelines by ingesting PDF, Word, HTML, and CSV documents, generating embeddings, and indexing knowledge in ChromaDB.

DELL TECHNOLOGIES | MACHINE LEARNING ENGINEER / APPLICATION ARCHITECT

Jul 2015 – May 2022 | Austin, TX

- Designed and implemented end-to-end Machine Learning pipelines using Python, Scikit-learn, Pandas, and FastAPI for predictive analytics and automated business decision support.
- Developed and optimized Logistic Regression models through feature engineering, model evaluation, and hyperparameter tuning to predict project feasibility with high accuracy and explainability.

EARLIER EXPERIENCE | IBM, INNOMINDS, AND WELLS FARGO

- Delivered enterprise-scale solutions in eCommerce, financial services, PLM systems, and workflow automation using Java technologies, SOA, Spring, and web services.

KEY ACHIEVEMENTS

- Designed and delivered enterprise AI solutions spanning Machine Learning, Generative AI, Retrieval-Augmented Generation (RAG), and Agentic AI, automating complex business workflows.
- Built production-grade multi-agent AI platforms using LangGraph, FastAPI, ChromaDB, Ollama, and LLMs for intelligent search, root cause analysis, recommendations, and workflow automation.